

- 29 Two radioactive isotopes P and Q have half-lives of 10 minutes and 15 minutes respectively. A sample initially contains the same number of atoms of each isotope. After 30 minutes, the ratio of the number of atoms of P to the number of atoms of Q will be

A 0.25

B 0.50

C 1.0

D 2.0

$$\frac{\text{the number of atoms of P}}{\text{the number of atoms of Q}} = \frac{\left(\frac{1}{2}\right)^{30/10}}{\left(\frac{1}{2}\right)^{30/15}} = \frac{\left(\frac{1}{2}\right)^3}{\left(\frac{1}{2}\right)^2} = \frac{1}{2} = 0.5$$

30 At time t a sample of a radioactive substance contains N atoms of a particular nuclide